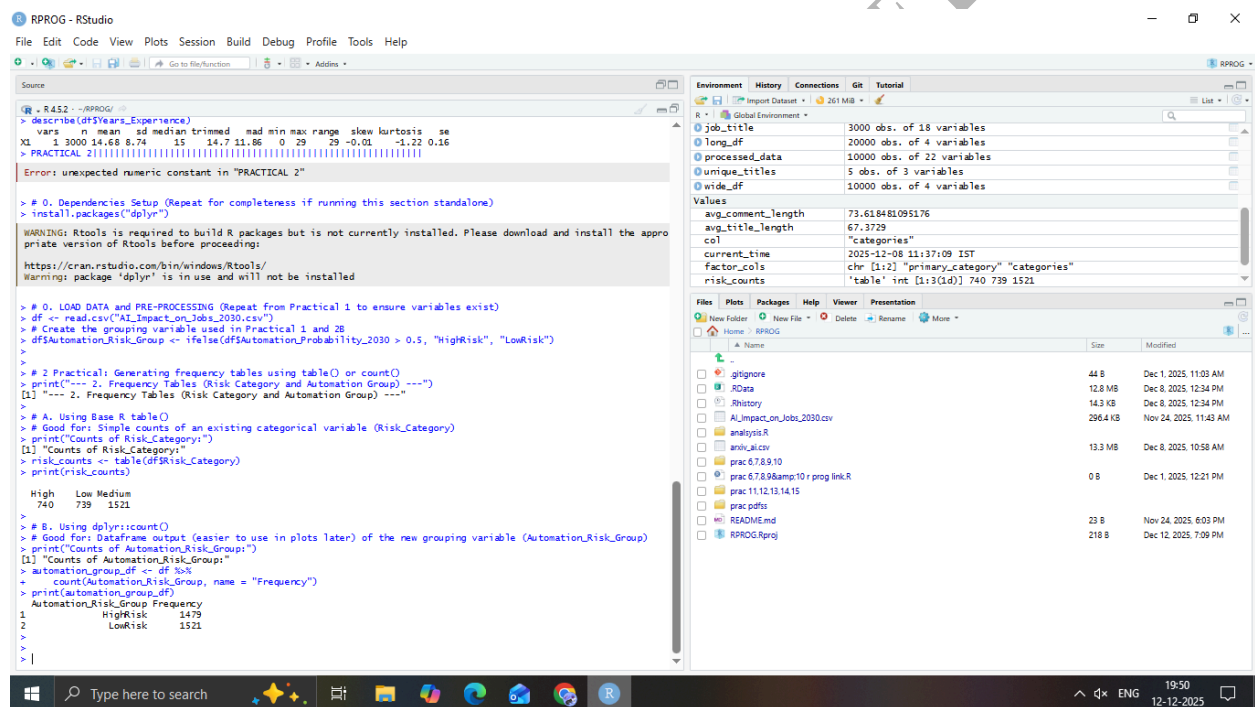


SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME : DATA ANALYSIS WITH SAS/SPSS/R

PRACTICAL 2

AIM: Generating frequency tables using table() or count()



The screenshot displays the RStudio environment. The Source pane on the left contains R code for data analysis. The Environment pane on the right shows the loaded data objects.

```
> R452 - ~/RPROG/
> describe(df$Years_Experience)
> vars  n mean sd median trimmed mad min max range skew kurtosis se
X1 1 3000 14.68 8.74 15 14.7 11.86 0 29 29 -0.01 -1.22 0.16
> PRACTICAL 2|||||
Error: unexpected numeric constant in "PRACTICAL 2"

> # 0. Dependencies Setup (Repeat for completeness if running this section standalone)
> install.packages("dplyr")

WARNING: Rtools is required to build R packages but is not currently installed. Please download and install the appropriate version of Rtools before proceeding:
https://cran.rstudio.com/bin/windows/Rtools/
Warning: package 'dplyr' is in use and will not be installed

> # 0. LOAD DATA AND PRE-PROCESSING (Repeat from Practical 1 to ensure variables exist)
> df <- read.csv("AI_Impact_on_Jobs_2030.csv")
> # Create the grouping variable used in Practical 1 and 2B
> df$Automation_Risk_Group <- ifelse(df$Automation_Probability_2030 > 0.5, "HighRisk", "LowRisk")
>
> # 2 Practical: Generating frequency tables using table() or count()
> print("---- 2. Frequency Tables (Risk Category and Automation Group) ----")
[1] "---- 2. Frequency Tables (Risk Category and Automation Group) ----"
>
> # A. Using Base R table()
> # Good for: Simple counts of an existing categorical variable (Risk_Category)
> print("Counts of Risk_Category:")
[1] "Counts of Risk_Category:"
> risk_counts <- table(df$Risk_Category)
> print(risk_counts)

High Low Medium
740 739 1521

> # B. Using dplyr::count()
> # Good for: Dataframe output (easier to use in plots later) of the new grouping variable (Automation_Risk_Group)
> print("Counts of Automation_Risk_Group:")
[1] "Counts of Automation_Risk_Group:"
> automation_group_df <- df %>%
+   count(Automation_Risk_Group, name = "Frequency")
> print(automation_group_df)
Automation_Risk_Group Frequency
1 HighRisk 1479
2 LowRisk 1521
>
> |
```

The Environment pane on the right shows the following objects:

Object	Details
job_title	3000 obs. of 18 variables
long_df	20000 obs. of 4 variables
processed_data	10000 obs. of 22 variables
unique_titles	5 obs. of 3 variables
wide_df	10000 obs. of 4 variables

The Values pane shows the following data:

Variable	Value
avg_comment_length	73.618461095176
avg_title_length	67.3729
col	"categories"
current_time	2025-12-08 11:37:09 IST
factor_cols	chr [1:2] "primary_category" "categories"
risk_counts	'table' int [1:3(d)] 740 739 1521

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SUBJECT NAME : DATA ANALYSIS WITH SAS/SPSS/R

The screenshot shows the RStudio interface with the following components:

- Source:** Displays the R version 4.5.2 (2025-10-31 ucrt) -- "[Not] Part in a Runtime" and the R Foundation for Statistical Computing. It also shows the R license and the installation of the 'dplyr' package. The console output indicates that 'dplyr' was successfully installed and loaded.
- Environment:** Lists the loaded packages: 'dplyr', 'R', 'RStudio', 'R6', 'Rcpp', 'RcppEigen', 'RcppArmadillo', 'RcppParallel', 'RcppProgress', 'RcppShiny', 'RcppUnit', 'RcppVisualizer', 'RcppX11', 'RcppX11R', 'RcppX11R2', 'RcppX11R3', 'RcppX11R4', 'RcppX11R5', 'RcppX11R6', 'RcppX11R7', 'RcppX11R8', 'RcppX11R9', 'RcppX11R10', 'RcppX11R11', 'RcppX11R12', 'RcppX11R13', 'RcppX11R14', 'RcppX11R15', 'RcppX11R16', 'RcppX11R17', 'RcppX11R18', 'RcppX11R19', 'RcppX11R20', 'RcppX11R21', 'RcppX11R22', 'RcppX11R23', 'RcppX11R24', 'RcppX11R25', 'RcppX11R26', 'RcppX11R27', 'RcppX11R28', 'RcppX11R29', 'RcppX11R30', 'RcppX11R31', 'RcppX11R32', 'RcppX11R33', 'RcppX11R34', 'RcppX11R35', 'RcppX11R36', 'RcppX11R37', 'RcppX11R38', 'RcppX11R39', 'RcppX11R40', 'RcppX11R41', 'RcppX11R42', 'RcppX11R43', 'RcppX11R44', 'RcppX11R45', 'RcppX11R46', 'RcppX11R47', 'RcppX11R48', 'RcppX11R49', 'RcppX11R50', 'RcppX11R51', 'RcppX11R52', 'RcppX11R53', 'RcppX11R54', 'RcppX11R55', 'RcppX11R56', 'RcppX11R57', 'RcppX11R58', 'RcppX11R59', 'RcppX11R60', 'RcppX11R61', 'RcppX11R62', 'RcppX11R63', 'RcppX11R64', 'RcppX11R65', 'RcppX11R66', 'RcppX11R67', 'RcppX11R68', 'RcppX11R69', 'RcppX11R70', 'RcppX11R71', 'RcppX11R72', 'RcppX11R73', 'RcppX11R74', 'RcppX11R75', 'RcppX11R76', 'RcppX11R77', 'RcppX11R78', 'RcppX11R79', 'RcppX11R80', 'RcppX11R81', 'RcppX11R82', 'RcppX11R83', 'RcppX11R84', 'RcppX11R85', 'RcppX11R86', 'RcppX11R87', 'RcppX11R88', 'RcppX11R89', 'RcppX11R90', 'RcppX11R91', 'RcppX11R92', 'RcppX11R93', 'RcppX11R94', 'RcppX11R95', 'RcppX11R96', 'RcppX11R97', 'RcppX11R98', 'RcppX11R99', 'RcppX11R100'.
- Files:** Shows the file structure of the project, including the 'R' folder and the 'Rproj' file.
- Plots:** Empty.
- Packages:** Shows the installed packages: 'dplyr', 'R', 'RStudio', 'R6', 'Rcpp', 'RcppEigen', 'RcppArmadillo', 'RcppParallel', 'RcppProgress', 'RcppShiny', 'RcppUnit', 'RcppVisualizer', 'RcppX11', 'RcppX11R', 'RcppX11R2', 'RcppX11R3', 'RcppX11R4', 'RcppX11R5', 'RcppX11R6', 'RcppX11R7', 'RcppX11R8', 'RcppX11R9', 'RcppX11R10', 'RcppX11R11', 'RcppX11R12', 'RcppX11R13', 'RcppX11R14', 'RcppX11R15', 'RcppX11R16', 'RcppX11R17', 'RcppX11R18', 'RcppX11R19', 'RcppX11R20', 'RcppX11R21', 'RcppX11R22', 'RcppX11R23', 'RcppX11R24', 'RcppX11R25', 'RcppX11R26', 'RcppX11R27', 'RcppX11R28', 'RcppX11R29', 'RcppX11R30', 'RcppX11R31', 'RcppX11R32', 'RcppX11R33', 'RcppX11R34', 'RcppX11R35', 'RcppX11R36', 'RcppX11R37', 'RcppX11R38', 'RcppX11R39', 'RcppX11R40', 'RcppX11R41', 'RcppX11R42', 'RcppX11R43', 'RcppX11R44', 'RcppX11R45', 'RcppX11R46', 'RcppX11R47', 'RcppX11R48', 'RcppX11R49', 'RcppX11R50', 'RcppX11R51', 'RcppX11R52', 'RcppX11R53', 'RcppX11R54', 'RcppX11R55', 'RcppX11R56', 'RcppX11R57', 'RcppX11R58', 'RcppX11R59', 'RcppX11R60', 'RcppX11R61', 'RcppX11R62', 'RcppX11R63', 'RcppX11R64', 'RcppX11R65', 'RcppX11R66', 'RcppX11R67', 'RcppX11R68', 'RcppX11R69', 'RcppX11R70', 'RcppX11R71', 'RcppX11R72', 'RcppX11R73', 'RcppX11R74', 'RcppX11R75', 'RcppX11R76', 'RcppX11R77', 'RcppX11R78', 'RcppX11R79', 'RcppX11R80', 'RcppX11R81', 'RcppX11R82', 'RcppX11R83', 'RcppX11R84', 'RcppX11R85', 'RcppX11R86', 'RcppX11R87', 'RcppX11R88', 'RcppX11R89', 'RcppX11R90', 'RcppX11R91', 'RcppX11R92', 'RcppX11R93', 'RcppX11R94', 'RcppX11R95', 'RcppX11R96', 'RcppX11R97', 'RcppX11R98', 'RcppX11R99', 'RcppX11R100'.
- Help:** Empty.
- Viewer:** Empty.
- Presentation:** Empty.

The screenshot shows the RStudio interface with the following components:

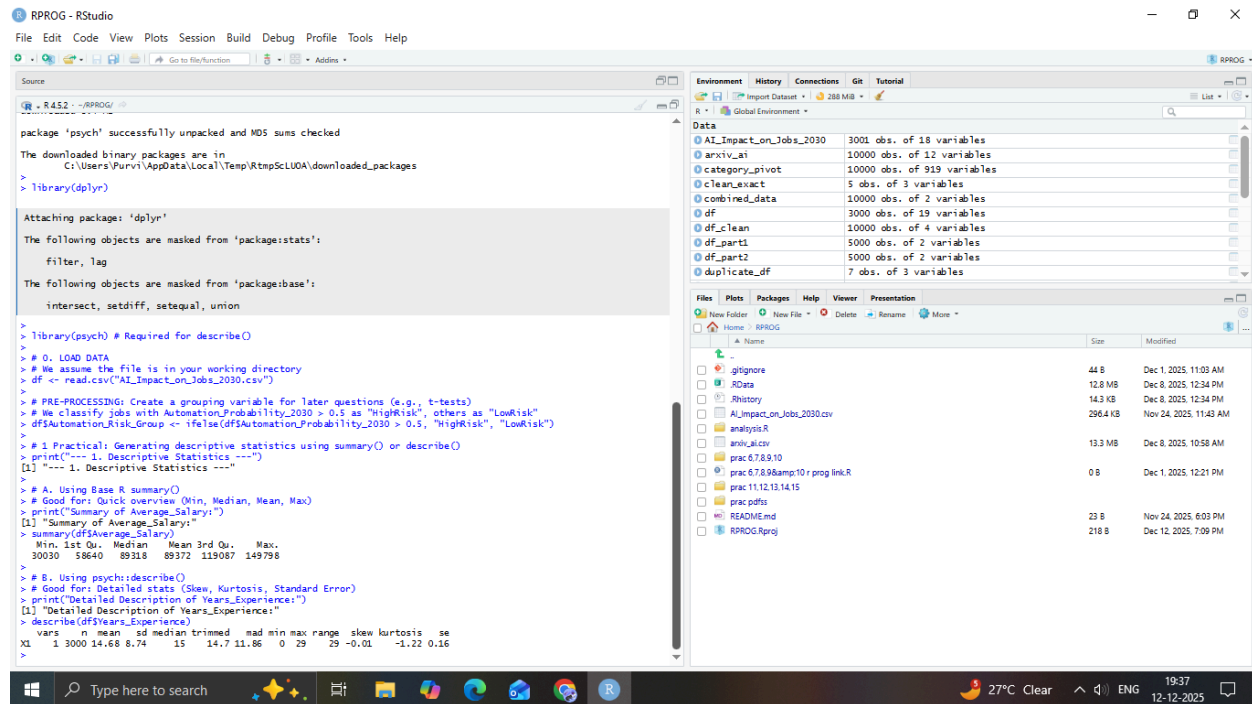
- Source:** Displays the R version 4.5.2 (2025-10-31 ucrt) -- "[Not] Part in a Runtime" and the R Foundation for Statistical Computing. It also shows the R license and the installation of the 'psych' package. The console output indicates that 'psych' was successfully installed and loaded.
- Environment:** Lists the loaded packages: 'dplyr', 'R', 'RStudio', 'R6', 'Rcpp', 'RcppEigen', 'RcppArmadillo', 'RcppParallel', 'RcppProgress', 'RcppShiny', 'RcppUnit', 'RcppVisualizer', 'RcppX11', 'RcppX11R', 'RcppX11R2', 'RcppX11R3', 'RcppX11R4', 'RcppX11R5', 'RcppX11R6', 'RcppX11R7', 'RcppX11R8', 'RcppX11R9', 'RcppX11R10', 'RcppX11R11', 'RcppX11R12', 'RcppX11R13', 'RcppX11R14', 'RcppX11R15', 'RcppX11R16', 'RcppX11R17', 'RcppX11R18', 'RcppX11R19', 'RcppX11R20', 'RcppX11R21', 'RcppX11R22', 'RcppX11R23', 'RcppX11R24', 'RcppX11R25', 'RcppX11R26', 'RcppX11R27', 'RcppX11R28', 'RcppX11R29', 'RcppX11R30', 'RcppX11R31', 'RcppX11R32', 'RcppX11R33', 'RcppX11R34', 'RcppX11R35', 'RcppX11R36', 'RcppX11R37', 'RcppX11R38', 'RcppX11R39', 'RcppX11R40', 'RcppX11R41', 'RcppX11R42', 'RcppX11R43', 'RcppX11R44', 'RcppX11R45', 'RcppX11R46', 'RcppX11R47', 'RcppX11R48', 'RcppX11R49', 'RcppX11R50', 'RcppX11R51', 'RcppX11R52', 'RcppX11R53', 'RcppX11R54', 'RcppX11R55', 'RcppX11R56', 'RcppX11R57', 'RcppX11R58', 'RcppX11R59', 'RcppX11R60', 'RcppX11R61', 'RcppX11R62', 'RcppX11R63', 'RcppX11R64', 'RcppX11R65', 'RcppX11R66', 'RcppX11R67', 'RcppX11R68', 'RcppX11R69', 'RcppX11R70', 'RcppX11R71', 'RcppX11R72', 'RcppX11R73', 'RcppX11R74', 'RcppX11R75', 'RcppX11R76', 'RcppX11R77', 'RcppX11R78', 'RcppX11R79', 'RcppX11R80', 'RcppX11R81', 'RcppX11R82', 'RcppX11R83', 'RcppX11R84', 'RcppX11R85', 'RcppX11R86', 'RcppX11R87', 'RcppX11R88', 'RcppX11R89', 'RcppX11R90', 'RcppX11R91', 'RcppX11R92', 'RcppX11R93', 'RcppX11R94', 'RcppX11R95', 'RcppX11R96', 'RcppX11R97', 'RcppX11R98', 'RcppX11R99', 'RcppX11R100'.
- Files:** Shows the file structure of the project, including the 'R' folder and the 'Rproj' file.
- Plots:** Empty.
- Packages:** Shows the installed packages: 'dplyr', 'R', 'RStudio', 'R6', 'Rcpp', 'RcppEigen', 'RcppArmadillo', 'RcppParallel', 'RcppProgress', 'RcppShiny', 'RcppUnit', 'RcppVisualizer', 'RcppX11', 'RcppX11R', 'RcppX11R2', 'RcppX11R3', 'RcppX11R4', 'RcppX11R5', 'RcppX11R6', 'RcppX11R7', 'RcppX11R8', 'RcppX11R9', 'RcppX11R10', 'RcppX11R11', 'RcppX11R12', 'RcppX11R13', 'RcppX11R14', 'RcppX11R15', 'RcppX11R16', 'RcppX11R17', 'RcppX11R18', 'RcppX11R19', 'RcppX11R20', 'RcppX11R21', 'RcppX11R22', 'RcppX11R23', 'RcppX11R24', 'RcppX11R25', 'RcppX11R26', 'RcppX11R27', 'RcppX11R28', 'RcppX11R29', 'RcppX11R30', 'RcppX11R31', 'RcppX11R32', 'RcppX11R33', 'RcppX11R34', 'RcppX11R35', 'RcppX11R36', 'RcppX11R37', 'RcppX11R38', 'RcppX11R39', 'RcppX11R40', 'RcppX11R41', 'RcppX11R42', 'RcppX11R43', 'RcppX11R44', 'RcppX11R45', 'RcppX11R46', 'RcppX11R47', 'RcppX11R48', 'RcppX11R49', 'RcppX11R50', 'RcppX11R51', 'RcppX11R52', 'RcppX11R53', 'RcppX11R54', 'RcppX11R55', 'RcppX11R56', 'RcppX11R57', 'RcppX11R58', 'RcppX11R59', 'RcppX11R60', 'RcppX11R61', 'RcppX11R62', 'RcppX11R63', 'RcppX11R64', 'RcppX11R65', 'RcppX11R66', 'RcppX11R67', 'RcppX11R68', 'RcppX11R69', 'RcppX11R70', 'RcppX11R71', 'RcppX11R72', 'RcppX11R73', 'RcppX11R74', 'RcppX11R75', 'RcppX11R76', 'RcppX11R77', 'RcppX11R78', 'RcppX11R79', 'RcppX11R80', 'RcppX11R81', 'RcppX11R82', 'RcppX11R83', 'RcppX11R84', 'RcppX11R85', 'RcppX11R86', 'RcppX11R87', 'RcppX11R88', 'RcppX11R89', 'RcppX11R90', 'RcppX11R91', 'RcppX11R92', 'RcppX11R93', 'RcppX11R94', 'RcppX11R95', 'RcppX11R96', 'RcppX11R97', 'RcppX11R98', 'RcppX11R99', 'RcppX11R100'.
- Help:** Empty.
- Viewer:** Empty.
- Presentation:** Empty.

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SUBJECT NAME : DATA ANALYSIS WITH SAS/SPSS/R



The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for loading the 'psych' package, reading a CSV file, and performing descriptive statistics. The code includes comments and function calls like `summary()` and `describe()`.
- Environment Pane:** Lists loaded data frames and their dimensions:
 - `AI_Impact_on_Jobs_2030`: 3001 obs. of 18 variables
 - `arxiv_ai`: 10000 obs. of 12 variables
 - `category_pivot`: 10000 obs. of 919 variables
 - `clean_exact`: 5 obs. of 3 variables
 - `combined_data`: 10000 obs. of 2 variables
 - `df`: 3000 obs. of 19 variables
 - `df_clean`: 10000 obs. of 4 variables
 - `df_part1`: 5000 obs. of 2 variables
 - `df_part2`: 5000 obs. of 2 variables
 - `duplicate_df`: 7 obs. of 3 variables
- Files Pane:** Shows the file explorer with various files and folders, including `AI_Impact_on_Jobs_2030.csv`.
- Console:** Displays the output of the R code, including package loading messages and the results of the `summary()` and `describe()` functions.

The console output for `summary(df$Average_Salary)` is as follows:

Wtn.	1st Qu.	Median	Mean	3rd Qu.	Max.
30030	58640	89318	89372	119087	149798

The console output for `describe(df$Years_Experience)` is as follows:

vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se	
X1	1	3000	14.68	8.74	15	14.7	11.86	0	29	29	-0.01	-1.22	0.16

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